Viper at BAREC Shared Task 2026: Structured Arabic Representations and Robust Multi-Seed Ensembling for Readability Assessment

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Abstract

Fine-grained Arabic readability assessment aims to estimate how difficult an Arabic sentence is for its intended readers by assigning it to one of 19 ordered readability levels. This task is particularly challenging because of Arabic's morphological complexity and the imbalanced distribution of examples across the levels. To address these challenges, we develop a system for the Sentence-Level Strict Track of the BAREC Shared Task 2026. Using only the official BAREC data, the system combines D3Tok-based morphological information, normalized surface text, and complementary features within a multi-seed ensemble of AraBERTv2-based regression models. Our submission achieved a QWK of 84.1% on the official Blind Test. This result indicates that the proposed combination provides competitive performance for fine-grained Arabic readability assessment.

1 Introduction

Fine-grained Arabic readability assessment assigns each sentence to one of 19 ordered difficulty levels. The task is challenging because Arabic words exhibit rich morphology, short-vowel diacritics are often omitted, and neighboring readability levels may differ only subtly. In addition, the BAREC label distribution is highly imbalanced, with substantially fewer examples at the easiest and hardest levels (Elmadani et al., 2025; Habash et al., 2025).

We present Team Viper's submission to the Sentence-Level Strict Track of BAREC-ST-2026. The system represents each sentence using paired D3Tok and normalized surface sequences, augmented with surface statistics, domain, and text-class information. We fine-tune an AraBERTv2-based regressor with imbalance-aware sampling and average the raw predictions of five independently trained models. Only the official BAREC training split is used for task-specific parameter optimization, while the development split is used for checkpoint selection. On the official Blind Test, our submission ranked fifth among 11 participating teams with a QWK of 84.1%.

2 Background

2.1 Task Definition

BAREC-ST-2026 defines sentence-level Arabic readability assessment as predicting an integer level in $\{1, \ldots, 19\}$ for each input sentence (Elmadani et al., 2026). Level 1 denotes the easiest sentences and Level 19 the most difficult. We participate in the Strict Track, where task-specific model training is restricted to the official BAREC training split. The development set is used for model selection, and the Blind Test is reserved for official evaluation. The primary metric is quadratic weighted kappa (QWK), which assigns larger penalties to disagreements farther apart on the ordinal scale.

2.2 Related Work

Pretrained language models have become central to readability assessment because they capture contextual lexical and syntactic information. BERT introduced deep contextual representations (Devlin et al., 2019), while AraBERT adapted this approach to Arabic (Antoun et al., 2020). For fine-grained Arabic readability, prior BAREC experiments showed that AraBERTv2 benefits from D3Tok input and that regression is effective for ordinal readability prediction (Elmadani et al., 2025). We generate D3Tok using CAMeL Tools (Obeid et al., 2020). Our system builds on these findings by combining D3Tok and surface inputs with structured features, imbalance-aware sampling, and multi-seed regression ensembling.

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 186–190
October 28-29, 2026 ©2026 Association for Computational Linguistics

3 System Overview

3.1 Structured Input Representation

For each sentence x, we construct two complementary input sequences. The first sequence contains its D3Tok morphological segmentation, generated using CAMeL Tools (Obeid et al., 2020). This representation separates relevant morphological components and makes the internal structure of Arabic words more explicit. The second sequence contains the normalized surface sentence together with lightweight statistics and corpus-provided metadata. Specifically, we include word count (`[WC]`), diacritic coverage (`[DC]`), average word length (`[WLA]`), word-length standard deviation (`[WLS]`), domain (`[DOM]`), and text class (`[TC]`).

The two sequences are jointly encoded as a paired input. This design retains the lexical information in the surface sentence while providing the encoder with additional cues related to morphology, sentence length, diacritization, domain, and text class. Together, these components provide complementary morphological, lexical, and corpus-level cues for distinguishing neighboring readability levels. Figure 1 summarizes the final paired input representation.

3.2 Regression and Imbalance-Aware Training

We initialize the encoder from the BAREC-specific `CAMeL-Lab/readability-arabertv2-d3tok-CE` checkpoint, which is based on AraBERTv2 (Antoun et al., 2020; Elmadani et al., 2025). The original classification head is discarded. Instead, the hidden representation of the `[CLS]` token is passed through dropout and a linear layer that produces a single continuous readability score $\hat{y}_i$.

We treat the ordered labels $y_i \in \{1, \ldots, 19\}$ as numerical targets and train the model using mean squared error (MSE):

$$\mathcal{L}_{\text{MSE}} = \frac{1}{B} \sum_{i=1}^{B} (\hat{y}_i - y_i)^2 ,$$

where B denotes the batch size. Unlike nominal classification, this formulation accounts for the distance between readability levels by assigning a larger loss to predictions farther from the reference label. Because the training labels are unevenly distributed, we apply weighted sampling. For an example i belonging to level y_i, its sampling weight is

$$w_i = \left(\frac{1}{n_{y_i}} \right)^{1/2} ,$$

where n_{y_i} is the number of training examples assigned to that level. This increases the sampling probability of underrepresented levels without excessive oversampling. The weighting affects only sample selection; the regression objective remains unchanged.

3.3 Multi-Seed Ensemble and Inference

We independently train five regressors using random seeds 42, 52, 62, 72, and 82. For each seed, we retain the checkpoint with the highest development-set QWK, using lower MAE to break ties. Let $f_m(x)$ denote the continuous prediction produced by model m. The ensemble score is the uniform average of the five raw predictions:

$$\bar{y}(x) = \frac{1}{5} \sum_{m=1}^{5} f_m(x).$$

The averaged score is converted to a valid readability level by rounding it to the nearest integer and clipping it to $[1, 19]$:

$$\hat{y}(x) = clip_{[1,19]} \left(round \left(\bar{y}(x) \right) \right) .$$

The ensemble uses neither learned model weights nor development-tuned thresholds. We use uniform averaging as a simple aggregation strategy, but do not isolate its contribution through a dedicated ablation study.

4 Experimental Setup

4.1 Data Protocol

We use the official sentence-level BAREC 2026 data splits. In accordance with the Sentence-Level Strict Track restrictions, only the official training split is used for parameter optimization. The development split is used for checkpoint selection and early stopping. Neither Open Test nor Blind Test information was used for checkpoint selection, threshold tuning, ensemble construction, or error analysis. The development set was used for checkpoint selection, early stopping, and post hoc error analysis in Section 5.1. No external corpora, synthetic examples, or pseudo-labels are used.

Before training, we validate the required identifier, sentence, and label fields; reject missing or

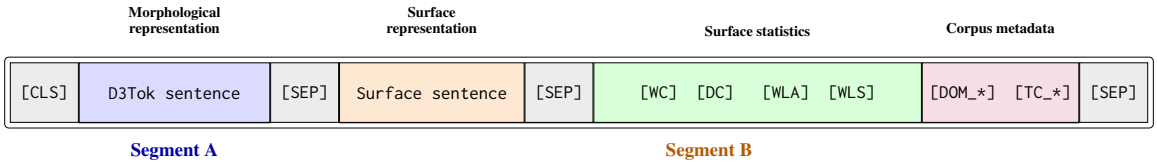


Figure 1: Structure of the combined model input. Segment A contains the D3Tok morphological representation. Segment B contains the normalized surface sentence, surface statistics, and corpus metadata. Special tokens delimit the individual components.

duplicate identifiers; and verify that all training and development labels are integers in the range $[1, 19]$. We additionally check that instance IDs and document IDs do not overlap across the official splits. The original input order is retained throughout preprocessing and inference to ensure that the final predictions follow the required submission order.

4.2 Implementation Details

D3Tok representations are generated using the MSA BERT unfactored disambiguator from CAMeL Tools (Obeid et al., 2020), with the highest-ranked analysis retained for each token. Unicode normalization is applied before morphological processing, while Kashida characters are removed without discarding Arabic diacritics required by the disambiguator. The resulting D3Tok and surface representations are cached separately for each data split to ensure that all ensemble members use identical preprocessed inputs. Table 1 summarizes the main training configuration. We use AdamW with separate learning rates for the pretrained encoder and the regression head. A linear learning-rate schedule is applied with warm-up over the first 10% of optimizer updates.

Hyperparameter	Value
Maximum sequence length	512
Training epochs	5
Per-device batch size	8
Encoder learning rate	2×10^{-5}
Regression-head learning rate	1×10^{-4}
Optimizer	AdamW
Warm-up ratio	0.10
Dropout	0.10
Early-stopping patience	2 epochs
Random seeds	42, 52, 62, 72, 82

Table 1: Training hyperparameters used for all ensemble members.

The system is implemented using PyTorch, Hugging Face Transformers, scikit-learn, and CAMeL Tools. For reproducibility, our source code, preprocessing pipeline, and training configuration are publicly available at https://github.com/POG42069/BAREC_SENT_STRICT_TASK.

4.3 Evaluation Protocol

We use QWK as the primary evaluation metric because the target labels are ordered and errors between distant levels should receive larger penalties than errors between adjacent levels. We also report exact 19-level accuracy (Acc_{19}), adjacent accuracy (± 1 Acc_{19}), and mean absolute error (MAE). Raw mean squared error (MSE) is used as a diagnostic measure for the continuous regression outputs.

5 Results

5.1 Official Leaderboard Performance

Table 2 presents the top eight systems on the official Testing Phase leaderboard for the Sentence-Level Strict Track. Our submission placed fifth with a QWK of 84.1%, 1.3 percentage points below the joint-best score of 85.4% and 0.1 percentage points behind the fourth-ranked system.

The fifth-place result indicates that our structured regression ensemble achieved competitive agreement with the ordered reference labels.

Rank	Participant	QWK
1	zaher-m	85.4
2	thylinao	85.4
3	Nile Nexus	84.8
4	jhliang	84.2
5	**Ours**	**84.1**
6	approx	83.8
7	phonghoang	83.7
8	ahmedabdou	83.6

Table 2: Top eight systems on the official Testing Phase leaderboard. QWK scores are reported as percentages.

5.2 Development Set Error Analysis

We aligned the ensemble predictions with the gold labels of all 7,310 development sentences using

their instance IDs. The system achieved a QWK of 84.3% and predicted the exact label for 3,005 sentences, corresponding to an exact accuracy of 41.1%. A total of 5,391 predictions (73.7%) were either correct or differed from the gold label by only one level. The mean absolute error was 1.15 levels.

Among the 4,305 incorrect predictions, 2,386 (55.4%) were off by exactly one level. Thus, more than half of the incorrect predictions fell immediately adjacent to the gold label. Although exact prediction across 19 readability levels remains challenging, nearly three quarters of all predictions were either correct or within one level of the gold label.

As shown in Figure 2, predictions are largely concentrated on or near the main diagonal. The most frequent misclassifications were 12→13 (291 sentences), 14→13 (269), 12→11 (220), 10→11 (197), and 14→15 (191). All five involve neighboring readability levels. This pattern shows that a substantial portion of the remaining errors occurs when distinguishing adjacent readability levels rather than levels that are far apart on the scale.

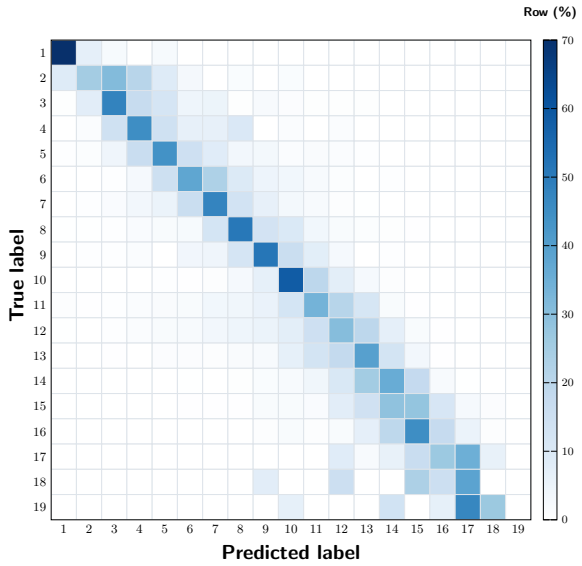


Figure 2: Confusion matrix on the development set.

Errors were particularly large at the upper end of the readability scale. Level 18 contained 13 development sentences and had a mean absolute error of 3.00 levels, while Level 19 contained 15 sentences and had a mean absolute error of 2.67. Neither level had an exact prediction. These results should be interpreted cautiously because both levels contain very few development examples, but they indicate that predictions at the rare upper levels remain challenging.

Performance also varied across metadata groups. Among the three domains, Social Sciences had the highest mean absolute error at 1.34 levels, compared with 1.15 for STEM and 1.09 for Arts & Humanities. Across text classes, Foundational texts had the highest mean absolute error at 1.22 levels, followed by Specialized texts at 1.15 and Advanced texts at 1.06. These results on the development set show that prediction difficulty varies across both domain and text class, with Social Sciences and Foundational texts showing the largest average prediction distances within their respective groupings.

6 Conclusion

We presented Team Viper's submission to the Sentence-Level Strict Track of BAREC-ST-2026. Our system combines D3Tok morphology, normalized surface text, lightweight structured features, imbalance-aware regression, and multi-seed ensembling. Using only the official BAREC data, our submission ranked fifth on the Blind Test leaderboard with a QWK of 84.1%. Analysis on the development set showed that most errors occurred between neighboring levels, while rare upper levels remained challenging. Social Sciences and Foundational texts showed the highest mean absolute errors within their respective metadata groupings. Future work will investigate more direct modeling of the ordinal label structure through differentiable QWK losses and ordinal regression, as well as calibrated decision boundaries in place of standard rounding.

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